NAME – VIVEK RAI

ROLL_NUMBER – 2401010303

SECTION – ‘B’

LAB_EXPERIMENT

PROJECT_TITLE-QUIZ WEBSITE

SUBMITTED_TO -**MANSI KAJAL MA’AM**

EXPERIMENT_03

HTML___CODE::::::::::::::

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Interactive Quiz</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="app" id="top">
    <header class="app-header">
      <div class="brand">
        <span class="brand-mark">IQ</span>
        <div class="brand-text">
          <h1>Interactive Quiz</h1>
          <p>Test your brain with quick multiple-choice questions.</p>
        </div>
      </div>
      <nav class="app-nav">
        <button class="nav-link" type="button" data-target="top">Home</button>
        <button class="nav-link" type="button" data-target="quiz">Quiz</button>
        <button class="nav-link" type="button" data-target="about">About</button>
      </nav>
    </header>
    <main class="app-main" id="quiz">
      <section class="quiz-card" aria-live="polite">
        <div class="quiz-top">
          <div class="progress">
            <div class="progress-bar">
              <div class="progress-fill"></div>
            </div>
            <span class="progress-text"></span>
          </div>
        </div>
        <section class="question-area">
          <h2 class="question-text"></h2>
          <div class="options" role="radiogroup"></div>
        </section>
        <section class="status-area">
          <p class="helper-text"></p>
        </section>
```

```
      <footer class="controls">
        <button class="btn ghost" id="restartBtn" type="button">Restart</button>
        <button class="btn primary" id="nextBtn" type="button">Next</button>
      </footer>
    </section>
    <section class="result-card" hidden>
      <div class="result-inner">
        <h2 class="result-title"></h2>
        <p class="result-score"></p>
        <p class="result-message"></p>
        <button class="btn primary" id="playAgainBtn" type="button">Play again</button>
      </div>
    </section>
  </main>
  <section class="about-card" id="about">
    <h2>About this quiz</h2>
    <p>This mini quiz helps you practice core web concepts in a fast, focused way.</p>
    <p>Each question gives instant visual feedback so your brain remembers better.</p>
  </section>
 </div>
 <script src="script.js"></script>
</body>
</html>
```

CSS CODE::::::

```
  color: #9ca3af;
}

.app-nav {
  display: flex;
  align-items: center;
  gap: 8px;
  flex-wrap: wrap;
}.about-card {
  background: rgba(15, 23, 42, 0.82);
  border-radius: 18px;
  padding: 18px 18px 20px;
  border: 1px solid rgba(148, 163, 184, 0.35);
  backdrop-filter: blur(16px);
  box-shadow: 0 18px 60px rgba(15, 23, 42, 0.9);
  display: grid;
  gap: 8px;
  font-size: 0.92rem;
  color: #d1d5db;
}
.app-header {
  flex-direction: column;
  align-items: flex-start;
}

.app-nav {
  width: 100%;
}

.nav-link {
  border: none;
  border-radius: 999px;
  padding: 6px 14px;
  font-size: 0.85rem;
  background: rgba(15, 23, 42, 0.9);
  color: #e5e7eb;
  border: 1px solid rgba(148, 163, 184, 0.5);
  cursor: pointer;
  transition: background 0.16s ease, border-color 0.16s
ease, transform 0.08s ease;
}

.nav-link:hover {
  background: rgba(59, 130, 246, 0.16);
  border-color: #60a5fa;
  transform: translateY(-1px);
}

body {
  min-height: 100vh;
  font-family: system-ui, -apple-system,
BlinkMacSystemFont, "Segoe UI", sans-serif;
  background: radial-gradient(circle at top left, #3b82f6,
transparent 50%), radial-gradient(circle at bottom right,
#ec4899, transparent 50%), #020617;
  color: #e5e7eb;
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 16px;
}

.app {
  width: 100%;
  max-width: 960px;
  display: grid;
  gap: 24px;
}

.app-header {
```

```
  text-align: center;
}

.app-header h1 {
  font-size: clamp(1.8rem, 3vw, 2.4rem);
  letter-spacing: 0.04em;
}

.app-header p {
  margin-top: 8px;
  color: #9ca3af;
}

.app-main {
  display: grid;
  grid-template-columns: minmax(0, 1.4fr) minmax(0, 1fr);
  gap: 20px;
}

.quiz-card,
.result-card {
  background: rgba(15, 23, 42, 0.82);
  border-radius: 18px;
  padding: 20px 18px 18px;
  border: 1px solid rgba(148, 163, 184, 0.35);
  backdrop-filter: blur(16px);
  box-shadow: 0 18px 60px rgba(15, 23, 42, 0.9);
}

.quiz-card {
  display: grid;
  grid-template-rows: auto 1fr auto auto;
  gap: 16px;
  position: relative;
  overflow: hidden;
}

.quiz-card::before {
  content: "";
  position: absolute;
  inset: 0;
  background: radial-gradient(circle at top, rgba(96, 165,
250, 0.18), transparent 60%);
  opacity: 0.8;
  pointer-events: none;
}

.quiz-card > * {
  position: relative;
  z-index: 1;
}

.quiz-top {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 12px;
}

.progress {
  width: 100%;
  display: flex;
  align-items: center;
  gap: 10px;
}

.progress-bar {
  flex: 1;
```

```
  height: 7px;
  border-radius: 999px;
  background: rgba(30, 64, 175, 0.5);
  overflow: hidden;
}

.progress-fill {
  height: 100%;
  width: 0;
  border-radius: inherit;
  background: linear-gradient(90deg, #38bdf8, #4f46e5, #f97316);
  transition: width 0.35s ease-out;
}

.progress-text {
  font-size: 0.8rem;
  color: #9ca3af;
  min-width: 60px;
  text-align: right;
}

.question-area {
  display: grid;
  gap: 14px;
}

.question-text {
  font-size: 1.1rem;
  font-weight: 600;
}

.options {
  display: grid;
  gap: 10px;
}

.option {
  display: flex;
  align-items: center;
  gap: 10px;
  padding: 10px 12px;
  border-radius: 999px;
  border: 1px solid rgba(148, 163, 184, 0.6);
  background: rgba(15, 23, 42, 0.9);
  cursor: pointer;
  transition: border-color 0.18s ease, background 0.18s ease, transform 0.08s ease;
}

.option span {
  font-size: 0.86rem;
  color: #9ca3af;
  min-width: 26px;
  text-align: center;
  border-radius: 999px;
  padding: 4px 8px;
  background: rgba(30, 64, 175, 0.55);
}

.option p {
  font-size: 0.95rem;
}

.option:hover {
  border-color: #60a5fa;
  transform: translateY(-1px);
}
```

```
.option.selected {
  border-color: #4f46e5;
  background: radial-gradient(circle at top left, rgba(129, 140, 248, 0.45), rgba(15, 23, 42, 0.95));
}

.option.correct {
  border-color: #22c55e;
}

.option.wrong {
  border-color: #ef4444;
}

.status-area {
  min-height: 20px;
}

.helper-text {
  font-size: 0.8rem;
  color: #9ca3af;
}

.controls {
  display: flex;
  justify-content: flex-end;
  gap: 10px;
  margin-top: 2px;
}

.btn {
  border-radius: 999px;
  border: none;
  padding: 9px 18px;
  font-size: 0.9rem;
  font-weight: 500;
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 8px;
  transition: transform 0.1s ease, box-shadow 0.1s ease, background 0.18s ease, color 0.18s ease;
}

.btn.primary {
  background: linear-gradient(135deg, #6366f1, #22d3ee);
  color: white;
  box-shadow: 0 10px 30px rgba(56, 189, 248, 0.45);
}

.btn.primary:disabled {
  opacity: 0.45;
  cursor: default;
  box-shadow: none;
}

.btn.ghost {
  background: transparent;
  color: #e5e7eb;
  border: 1px solid rgba(148, 163, 184, 0.6);
}

.btn:not(:disabled):hover {
  transform: translateY(-1px);
  box-shadow: 0 14px 34px rgba(15, 23, 42, 0.9);
}
```

```
.btn:not(:disabled):active {
  transform: translateY(0);
  box-shadow: none;
}

.result-card {
  display: flex;
  align-items: center;
  justify-content: center;
}

.result-inner {
  text-align: center;
  display: grid;
  gap: 10px;
}

.result-title {
  font-size: 1.4rem;
}

.result-score {
  font-size: 1.1rem;
  color: #a5b4fc;
}

.result-message {
  color: #9ca3af;
}

.result-card.show {
  animation: floatIn 0.4s ease-out;
}

.quiz-card.fade {
  animation: fadeIn 0.2s ease-out;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(4px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

@keyframes floatIn {
  from {
    opacity: 0;
    transform: translateY(10px) scale(0.97);
  }
  to {
    opacity: 1;
    transform: translateY(0) scale(1);
  }
}

@media (max-width: 768px) {
  .app-main {
    grid-template-columns: minmax(0, 1fr);
  }

  .quiz-card,
  .result-card {
```

```
    padding: 18px 16px 16px;
  }
}

@media (max-width: 480px) {
  body {
    padding: 10px;
  }

  .controls {
    justify-content: space-between;
  }

  .btn {
    flex: 1;
  }

  .option {
    align-items: flex-start;
  }
}
```

JS CIDE::

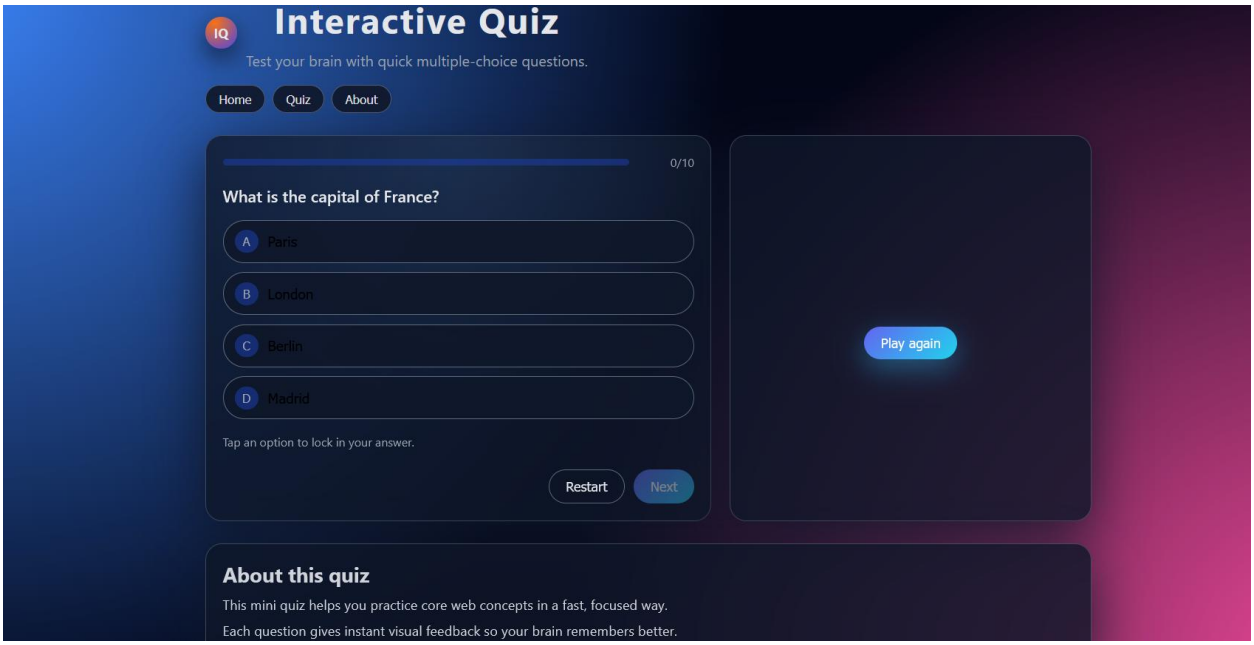
```
const questions = [
  {
    text: "What is the capital of France?",
    options: ["Paris", "London", "Berlin", "Madrid"],
    answer: 0
  },
  {
    text: "What is the capital of Germany?",
    options: ["Berlin", "London", "Madrid", "Paris"],
    answer: 0
  },
  {
    text: "What is the capital of Italy?",
    options: ["Rome", "London", "Madrid", "Paris"],
    answer: 0
  },
  {
    text: "What is the capital of Spain?",
    options: ["Madrid", "London", "Berlin", "Paris"],
    answer: 3
  },
  {
    text: "What is the capital of Portugal?",
    options: ["Lisbon", "London", "Berlin", "Paris"],
    answer: 0
  },
  {
    text: "What is the capital of Greece?",
    options: ["Athens", "London", "Berlin", "Paris"],
    answer: 0
  },
  {
    text: "What is the capital of Japan?",
    options: ["Tokyo", "London", "Berlin", "Paris"],
    answer: 0
  },
  {
    text: "What is the capital of China?",
    options: ["Beijing", "London", "Berlin", "Paris"],
    answer: 0
  },
  {
    text: "What is the capital of India?",
    options: ["New Delhi", "London", "Berlin", "Paris"],
    answer: 0
```

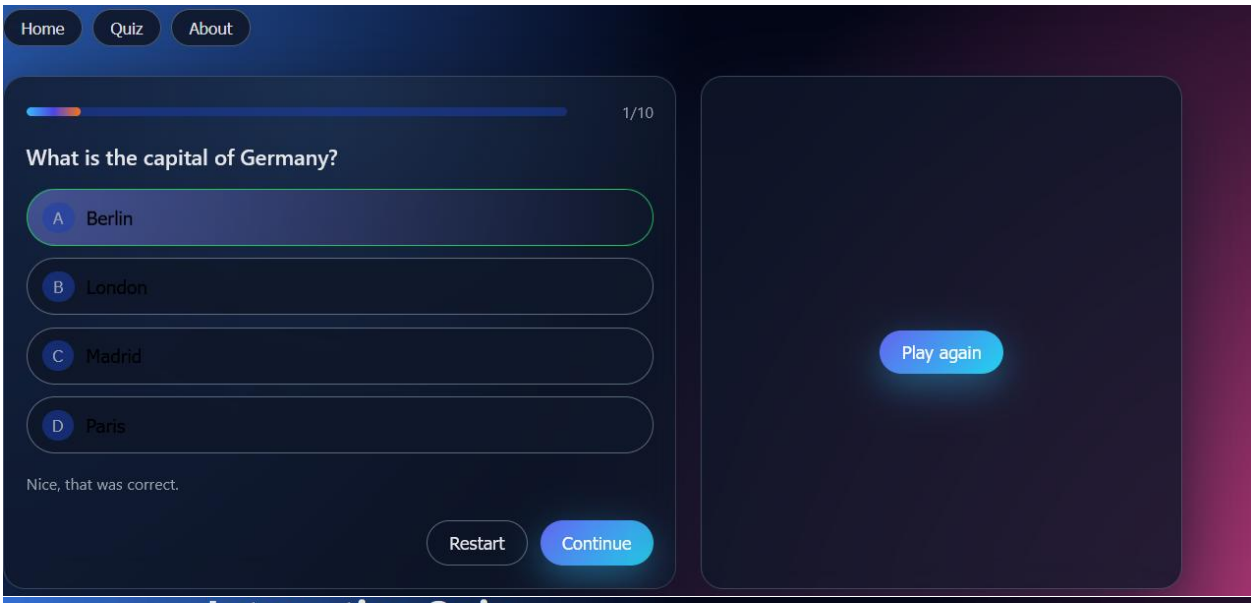
```
  },
  {
    text: "What is the capital of Brazil?",
    options: ["Brasilia", "London", "Berlin", "Paris"],
    answer: 0
  },




]


const questionText = document.querySelector(".question-text")
const optionsBox = document.querySelector(".options")
const helperText = document.querySelector(".helper-text")
const progressFill = document.querySelector(".progress-fill")
const progressText = document.querySelector(".progress-text")
const quizCard = document.querySelector(".quiz-card")
const resultCard = document.querySelector(".result-card")
const resultTitle = document.querySelector(".result-title")
const resultScore = document.querySelector(".result-score")
const resultMessage = document.querySelector(".result-message")
const nextBtn = document.getElementById("nextBtn")
const restartBtn = document.getElementById("restartBtn")
const playAgainBtn = document.getElementById("playAgainBtn")


let current = 0
let score = 0
let locked = false
let selectedIndex = null


function setProgress() {
  const step = (current / questions.length) * 100
  progressFill.style.width = `${Math.round(step)}%`
  progressText.textContent = `${current}/${questions.length}`
}


function renderQuestion() {
  const item = questions[current]
  quizCard.classList.remove("fade")
  void quizCard.offsetWidth
  quizCard.classList.add("fade")
  questionText.textContent = item.text
  optionsBox.innerHTML = ""
  helperText.textContent = "Tap an option to lock in your answer."
  locked = false
  selectedIndex = null
  nextBtn.disabled = true
  nextBtn.textContent = current === questions.length - 1 ? "Submit" : "Next"
  item.options.forEach((label, index) => {
    const btn = document.createElement("button")
    btn.className = "option"
    btn.type = "button"
    btn.setAttribute("role", "radio")
    btn.setAttribute("aria-checked", "false")
    btn.innerHTML = `<span>${String.fromCharCode(65 + index)}</span><p>${label}</p>`
    btn.addEventListener("click", () => handleSelect(index, btn))
    optionsBox.appendChild(btn)
  })
  setProgress()
}


function handleSelect(index, btn) {
```

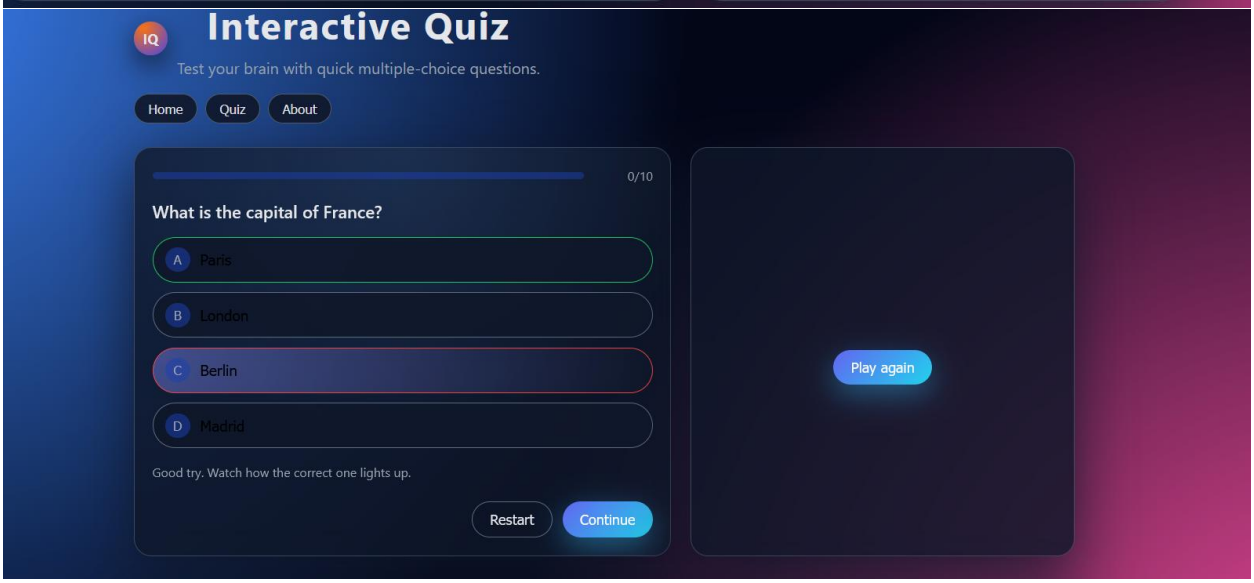
```
  if (locked) return

  const optionButtons = [...document.querySelectorAll(".option")]

  optionButtons.forEach(o => {

    o.classList.remove("selected")

    o.setAttribute("aria-checked", "false")

  })

  btn.classList.add("selected")

  btn.setAttribute("aria-checked", "true")

  selectedIndex = index

  helperText.textContent = "Ready? Hit next to check your answer."

  nextBtn.disabled = false

}


function revealAnswer() {

  const item = questions[current]

  const optionButtons = [...document.querySelectorAll(".option")]

  optionButtons.forEach((btn, index) => {

    if (index === item.answer) btn.classList.add("correct")

    if (index === selectedIndex && index !== item.answer) btn.classList.add("wrong")

  })

  if (selectedIndex === item.answer) {

    score++

    helperText.textContent = "Nice, that was correct."

  } else {

    helperText.textContent = "Good try. Watch how the correct one lights up."

  }

}


function handleNext() {

  if (!locked) {

    locked = true

    revealAnswer()

    nextBtn.textContent = current === questions.length - 1 ? "Finish" : "Continue"

    return

  }

  if (current < questions.length - 1) {

    current++

    renderQuestion()

  } else {

    showResult()

  }

}


function scoreMessage(ratio) {

  if (ratio === 1) return "Excellent, a perfect brain run."

  if (ratio >= 0.7) return "Great job, you are thinking like a developer."

  if (ratio >= 0.4) return "Good, a bit more practice will make it shine."

  return "Try again, every attempt wires your brain better."

}


function showResult() {

  const ratio = score / questions.length

  resultTitle.textContent = ratio >= 0.7 ? "Well played!" : "You can beat this."

  resultScore.textContent = `You scored ${score} / ${questions.length}`

  resultMessage.textContent = scoreMessage(ratio)

  quizCard.hidden = true

  resultCard.hidden = false

  resultCard.classList.add("show")

}


function resetQuiz() {

  current = 0
```

```
  score = 0

  locked = false

  selectedIndex = null

  resultCard.hidden = true

  quizCard.hidden = false

  helperText.textContent = "Tap an option to lock in your answer."

  restartBtn.style.opacity = "1"

  restartBtn.style.pointerEvents = "auto"

  renderQuestion()

}


const navLinks = document.querySelectorAll(".nav-link")


function scrollToSection(id) {

  const el = id === "quiz" ? quizCard : document.getElementById(id)

  if (el) el.scrollIntoView({ behavior: "smooth", block: "start" })

}


navLinks.forEach(btn => {

  btn.addEventListener("click", () => scrollToSection(btn.dataset.target))

})


nextBtn.addEventListener("click", handleNext)

restartBtn.addEventListener("click", resetQuiz)

playAgainBtn.addEventListener("click", resetQuiz)


resetQuiz()
```

SCREENSHOT OF WEBSITE----------




Home
Quiz
About
1/10
What is the capital of Germany?
A Berlin
B London
C Madrid
D Paris
Nice, that was correct.
Restart
Continue
Play again


IQ
Interactive Quiz
Test your brain with quick multiple-choice questions.
Home
Quiz
About
0/10
What is the capital of France?
A Paris
B London
C Berlin
D Madrid
Good try. Watch how the correct one lights up.
Restart
Continue
Play again